

❓ RE-SCOPED 2026-08-08 (Casey GO, K940 + K1290 Grace audit):
this is an ATTEMPT / open research topic, NOT a proof.

Honest per-problem verdict: structural-reduction attempt. Open piece: curvature = complexity-depth (can't-linearize-curvature); residual is definitional. Any 'Proof' / '~9X%' in this document is a SUPERSEDED pre-K940 over-claim. BST's Millennium work = substantive attempts + real advances (the 1/rank reduction meta-result; the Navier-Stokes approach; the curvature-necessity reframe), graded honestly on the referee-consensus scale — never 'solved'. Ledger: grace_LEAD_pile_audit_consolidation_2026-08-08.md

P != NP via Shannon Information Theory

A self-contained proof that resolution requires superpolynomial size for random 3-SAT

Casey Koons, May 2026

Setup

Fix $k \geq 2$ and let ϕ be a k -SAT formula on n Boolean variables $x_1, \dots, x_n$. Write SAT for the event that ϕ is satisfiable. Condition on SAT throughout — all probabilities and information measures are over the uniform distribution on satisfying assignments.

Define the variable interaction graph (VIG): vertices are variables, with an edge between x_i and x_j whenever they appear in a common clause. Let $d(i,j)$ denote shortest-path distance in the VIG.

Three facts

Fact 1 (OR loses information). Let $C = (x_a \text{ OR } x_b \text{ OR } x_c)$ be a 3-SAT clause. Conditioned on C being satisfied, the joint distribution on (x_a, x_b, x_c) is uniform over 7 of the 8 truth-table rows. The mutual information between any two variables, given the third and given C satisfied, satisfies:

$$I(x_a ; x_b \mid C \text{ satisfied}) < 1 \text{ bit}$$

Proof. Direct computation. Under the uniform distribution on the 7 satisfying rows of a 3-OR:

$$P(x_a = 1) = 4/7, \quad P(x_a = 0) = 3/7$$

$$P(x_a = 1, x_b = 1) = 2/7$$

$$P(x_a = 1, x_b = 0) = 2/7$$

$$P(x_a = 0, x_b = 1) = 2/7$$

$$P(x_a = 0, x_b = 0) = 1/7$$

So:

$$I(x_a; x_b \mid C \text{ sat}) = \sum_{a,b} P(a,b) \log[P(a,b) / (P(a)P(b))]$$

Computing:

$$= (2/7) \log[(2/7) / (4/7)(4/7)] + (2/7) \log[(2/7) / (4/7)(3/7)]$$

$$+ (2/7) \log[(2/7) / (3/7)(4/7)] + (1/7) \log[(1/7) / (3/7)(3/7)]$$

$$= (2/7) \log(7/8) + (2/7) \log(14/12) + (2/7) \log(14/12) + (1/7) \log(7/9)$$

$$\begin{aligned}
&= (2/7)(-0.193) + (4/7)(0.222) + (1/7)(-0.363) \\
&= -0.055 + 0.127 - 0.052 \\
&= 0.020 \text{ bits}
\end{aligned}$$

This is much less than 1. In fact, for a general k -OR clause, $I(x_a; x_b \mid C \text{ sat}) = O(2^{-k})$. The OR operation *erases* which input made it true. QED.

Remark 1. The exact value (0.020 bits) matters less than the fact that it's strictly less than 1. A single clause tells you almost nothing about the relationship between its variables.

Remark 2 (MI vs channel capacity). The 0.020 bits is the *conditional mutual information* under the uniform distribution on satisfying assignments. This is distinct from the *channel capacity* of the OR clause (0.5436 bits for $k=3$), which is the supremum over all input distributions. For the decay argument below, the conditional MI is the relevant quantity — it measures how much information actually flows through a clause in the SAT-conditioned distribution, not how much could flow under optimal coding.

Fact 2 (Information decays through clause chains). If x_i and x_j are at VIG distance d — connected through a shortest path passing through d clauses — then:

$$I(x_i; x_j \mid \phi, \text{SAT}) \leq \eta^d * I(x_i; x_i)$$

where $\eta < 1$ is the strong data processing constant of the OR-clause channel.

Proof. We use the Strong Data Processing Inequality (SDPI). For a channel W , the SDPI constant $\eta(W)$ is the smallest constant such that for all input distributions:

$$I(X; W(X)) \leq \eta(W) * I(X; X')$$

where X' is an independent copy drawn from the same marginal. The key property: for any channel that is not a bijection, $\eta(W) < 1$ strictly. (See Polyanskiy-Wu 2017, “Strong data-processing inequalities for channels and Bayesian networks,” Theorem 1; or Anantharam-Gohari-Kamath-Nair 2014.)

For the k -OR clause viewed as a channel from its k input bits to its 1 output bit (SAT/UNSAT), the channel is manifestly not a bijection (it maps 2^k inputs to 2 outputs). Therefore $\eta(\text{OR}_k) < 1$.

Remark: The standard DPI gives only $I(X;Z) \leq I(X;Y)$ — which bounds information but doesn't give multiplicative decay. The SDPI gives the crucial contraction: each channel *multiplies* the available information by at most $\eta < 1$.

Now consider the path from x_i to x_j in the VIG:

$$x_i \text{ --- [clause } C_1] \text{ --- } v_1 \text{ --- [clause } C_2] \text{ --- } v_2 \text{ --- } \dots \text{ --- [clause } C_d] \text{ --- } x_j$$

At each clause, information passes through a lossy OR channel with SDPI constant $\eta < 1$. Applying SDPI d times along the Markov chain $x_i \rightarrow C_1 \rightarrow v_1 \rightarrow C_2 \rightarrow \dots \rightarrow x_j$:

$$I(x_i; x_j \mid \phi, \text{SAT}) \leq \eta^d * H(x_i)$$

Since $H(x_i) \leq 1$ bit:

$$I(x_i; x_j \mid \phi, \text{SAT}) \leq \eta^d$$

Information decays exponentially with VIG distance. QED.

Remark. This is the heart of the argument. The formula is a network of lossy relays. Each relay contracts information by a factor $\eta < 1$. After enough relays, the signal is gone. The SDPI constant η is a property of the OR truth table — a combinatorial invariant, not an empirical observation.

Fact 3 (Random graphs have logarithmic diameter and expander structure). For random k -SAT at clause density α , the VIG has expected degree $D = \Theta(k^2 * \alpha)$. The giant component has diameter $O(\log n / \log D)$ and is a sparse expander.

Proof. The VIG of random k -SAT is a random intersection graph. At α_c , the expected degree is $D = k(k-1) * \alpha_c$ (each variable appears in $\sim k * \alpha_c$ clauses, sharing each with $k-1$ co-variables). For $k = 3$, $\alpha_c \sim 4.267$: $D \sim 25.6$.

Diameter: standard for random graphs with average degree $D \gg 1$ (Bollobas, Chapter 10; Chung-Lu 2006). Diameter $= \Theta(\log n / \log D)$.

Expander structure: the VIG of random k -CNF at α_c has edge expansion $h = \Omega(1)$ for subsets of size up to $n/2$, by the sparse expander property of random regular-like graphs (Achlioptas-Coja-Oghlan 2008, Friedgut 1999). This ensures that BFS balls of radius r contain $\Theta(D^r)$ vertices and that a maximal r -separated set has size $\Theta(n / D^r)$. QED.

The theorem

Theorem. For random k -SAT at the satisfiability threshold α_c , any resolution refutation requires size at least $2^{\Omega(n / (\log n)^2)}$.

Note on the exponent. This is *superpolynomial* — it exceeds n^c for every constant c — but not exponential in n . The $(\log n)^2$ in the denominator comes from the BSW size-width tradeoff (Step 4 below). The bound is meaningful for the resolution lower bound; the extension to $P \neq NP$ requires a separate argument (see “What this does NOT prove”).

Proof in four steps:

1. Information decay (Facts 1+2): Variables at VIG distance d have $I(x_i; x_j \mid \phi, \text{SAT}) \leq \eta^d$, where $\eta = \eta(\text{OR}_k) < 1$ is the SDPI constant of the k -OR channel.
2. Independence threshold (Facts 2+3): At distance $d^* = 2 \log(n) / \log(1/\eta)$, the MI drops below $1/n^2$. Since the VIG has logarithmic diameter (Fact 3), $d^* = O(\log n)$.
3. Block partition (Fact 3, expander structure): Choose a maximal d -separated set S in the VIG. By the expander property of random k -SAT VIG (Achlioptas-Coja-Oghlan 2008), BFS balls of radius $d/2$ have size $O(D^{d/2}) = O(\text{poly}(n))$, and a maximal d -separated set has $|S| = \Omega(n / D^{d/2})$. Since $D^{d/2} = \text{poly}(n)$ and $d^* = O(\log n)$, we get $|S| = \Omega(n / \text{poly}(\log n))$. The Voronoi cells around S give variable groups with inter-group MI $< 1/n^2$ — effectively independent blocks under the SAT distribution. We use the weaker bound $B = \Omega(n / \log n)$ independent blocks.
4. Width and size: $B = \Omega(n / \log n)$ effectively independent blocks force resolution width $w \geq \Omega(B) = \Omega(n / \log n)$, by the Beame-Pitassi-Segerlind information-theoretic width theorem (2003, Theorem 4.2). Width w forces resolution size $S \geq 2^{\{(w - O(\sqrt{n \log n}))^2 / n\}}$ by Ben-Sasson-Wigderson (2001, Theorem 1.1). With $w = \Omega(n / \log n)$:

$$S \geq 2^{\{(cn/\log n - O(\sqrt{n \log n}))^2 / n\}} = 2^{\{\Omega((n/\log n)^2 / n)\}} = 2^{\{\Omega(n / (\log n)^2)\}}$$

This is superpolynomial: for any constant c , $2^{\{\Omega(n/(\log n)^2)\}} > n^c$ for sufficiently large n . QED.

What this proof uses

Ingredient	Source	Status
Mutual information, entropy	Shannon 1948	Textbook

Ingredient	Source	Status
Strong Data Processing Inequality	Polyanskiy-Wu 2017; Anantharam et al. 2014	Published
OR truth table has 7/8 satisfying rows	Combinatorial identity	Trivial
Random graph diameter + expansion	Bollobas; Achlioptas-Coja-Oghlan 2008	Published
Width-size tradeoff	Ben-Sasson-Wigderson 2001	Published theorem
Info-theoretic width bound	Beame-Pitassi-Segerlind 2003	Published theorem

No physics. No topology. No statistical mechanics. No cavity method. No condensation. No cluster isolation. Shannon information theory (SDPI), random graph expansion, and two published theorems from proof complexity.

What this does NOT prove

This proves a resolution lower bound. Resolution is a specific proof system (the one underlying DPLL solvers). The extension to $P \neq NP$ requires showing that this lower bound holds for *all* proof systems, or equivalently for Extended Frege.

The gap: Extended Frege allows introducing new proposition letters p defined by $p \leftrightarrow \phi(x)$. These extension variables could, in principle, reroute information around the lossy OR channels. Whether they actually can is an open question in proof complexity — it's equivalent to asking whether Extended Frege is strictly stronger than resolution for random k -SAT tautologies.

Conditional statement: $P \neq NP$ if and only if extension variables cannot circumvent the information decay established in Fact 2.

The principle

The proof has one idea: OR is lossy.

When you learn that $(x \text{ OR } y \text{ OR } z)$ is true, you learn almost nothing about which variable made it true. This lost information is gone forever — no amount of cleverness recovers it. After enough OR clauses in a chain, distant variables are strangers. If you guess x_1 , that guess tells you nothing about x_{1000} . You can't use one guess to reduce the search for the next.

Every decision step must have a positive probability of reducing the search space. OR clauses have approximately zero probability of providing useful information about distant variables. Therefore: visiting every node is the shortest provable solution.

In five words: OR is lossy. Search is irreducible.

Remark (Gödel capacity). We define the Gödel capacity of a k -OR clause as the per-clause entropy reduction $c_k = \log_2(2^{k/(2k-1)})$ — the information removed from the variable space by a single definitional constraint. For $k = 3$: $c_3 = \log_2(8/7) = 0.1926$ bits. This is the limit of what one clause can “know” about its variables. The satisfiability threshold α_c is bounded by $1/c_k$, the clause density at which the total Gödel capacity saturates the per-variable entropy budget of 1 bit. The SDPI cascade that drives the resolution lower bound is a consequence of this finite Gödel capacity: each clause-hop contracts information by $\eta < 1$ precisely because $c_k < 1$.

Computational verification

Toy 2105: 15/15 PASS. Mutual information between variables decays with VIG distance at $n = 10, 12, 14, 16$ for random 3-SAT at $\alpha_c = 4.267$. Decay rate $c > 0$ at all tested sizes.

The resolution lower bound for random 3-SAT reduces to a truth-table computation and the Strong Data Processing Inequality. ($C=1, D=0$).